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Phase-dependent thermophysical properties of α-and γ-Al_2O_3 in aqueous suspension

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ABSTRACT

This study demonstrates the thermal conductivity (TC) and viscosity of as prepared crystalline α-Al_2O_3 and amorphous γ-Al_2O_3 particles, having size in the range of 30–50 nm. The α and γ-Al_2O_3 aqueous suspension exhibited ~10% and 6% enhancement in TC than de-ionized water, but α-Al_2O_3 showed (~4–6%) higher TC than γ-Al_2O_3 aqueous suspension due to more crystallinity of α phase than γ phase. Ultrasonication helps in the breakdown of large clusters which further improves the dispersion stability and TC as verified by dynamic light scattering and zeta potential measurements. The Al_2O_3 aqueous suspension showed Newtonian characteristics at lower concentration.
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Introduction

Nanofluid (NF), a suspension of nanoparticles in a base liquid, has been found to provide a considerable heat-transfer enhancement compared to "conventional" fluids such as water and ethylene glycol, the suspended crystalline nanoparticles have three orders of thermal conductivities (TC) larger than that of the base liquid [1–4]. Due to continuous growth in technology and upcoming more powerful tools in optics, lasers, communication, and electronics having miniaturized size, cooling has become an important aspect for thermal [5,6] management. The aqueous or non-aqueous suspension of Al_2O_3 nanoparticles (NPs) is recently studied for cooling applications due to high TC of solid NPs than liquid [7–9]. Particles with a nanoscale dimension provide a more surface area, which not only enhances the heat transfer rate but also increases the particles suspension stability in the conventional heat transfer fluids [10,11]. Researchers have been trying to enhance the heat transfer performance of conventional coolants by suspending tiny metallic, non-metallic and metal oxide particles in DI water and ethylene glycol (EG) depending upon their volume fraction (VF), particle size, shape, pH and temperature [12–15]. Some research groups [16–19] found 4 to 30% enhancement in TC by varying particle size (10 to 200 nm) and volume fraction (0.05 to 4 vol.%) of γ-Al_2O_3 in DI water and EG. Beck et al. [20] showed decrease in TC of alumina nanoparticles dispersion having size lower than 50 nm and TC was increased with increasing particle sizes (100–282 nm). Teng et al. [21] presented the effect of particle size, concentration and temperature on increased TC of aqueous suspension of Al_2O_3 particles. In some other reports [22–24] found that by increasing the temperature from 31 to 51 °C, the enhancement in TC was found to be 2 to 32.4% for Al_2O_3 dispersed in DI water and EG. Kole et al. [25] dispersed <50-nm Al_2O_3 (3.5 vol.%) using oleic acid as surfactant in a car engine coolant and observed 10.41% TC enhancement and demonstrated the stability of such fluids for more than 80 days. It may be mentioned here that engineering applications of NPs suspension that employ fluid flow not only require information on their thermal properties, but the appropriate rheological properties of the aqueous suspension is also very significant [26]. In the Al_2O_3–water mixture viscosity (VS) increases between 20% and 30% for 3 vol.% Al_2O_3 solution as compared to water alone [27]. Das et al. [28] demonstrated an increase of VS with increased particle concentrations and Kole et al. [25] demonstrated a transition from Newtonian characteristics to non-Newtonian behavior with increasing content of Al_2O_3 in the engine coolant. Recently, Beck et al. [29] suggested the temperature dependence TC and VS of EG-based suspension in the range of 296 to 400 K. Chandrasekar et al. [30] observed Nusselt numbers were increased by 15.91% and 21.53% when Al_2O_3/water nanofluid is used with wire coil inserts WC2 and WC3, respectively, at Re = 2275 compared to those of

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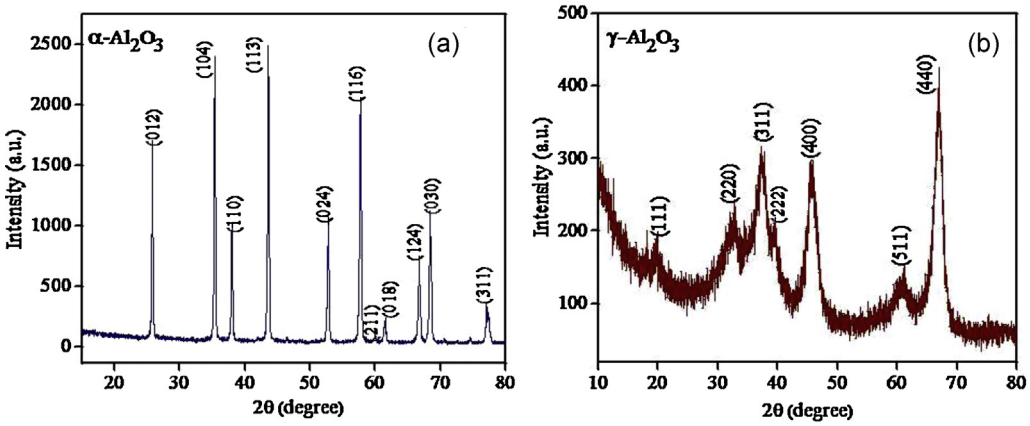


Fig. 1. XRD patterns of (a) α-Al_2O_3 and (b) γ-Al_2O_3 particles.

distilled water. Teng et al. [31] also studied the effect of phase transition of Al_2O_3 on thermal conductivity.

Alumina exists in a variety of metastable structures, such as χ, κ, γ, η, δ, θ, and ρ, as well as its stable α-Al_2O_3 phase and each has a unique crystal structure and properties [32,33]. This polymorphism can be categorized in terms of the oxygen sublattice structure and the distribution into this sublattice of aluminum ions in tetrahedral and octahedral interstitial sites [34]. Thus, in α-Al_2O_3, the oxygen sublattice is hexagonal-close-packed (hcp) structured with 2/3 of octahedral sites occupied with cations, while γ, η, θ have a face-centerd cubic (fcc) array of oxygen atoms and cations present in a variety of proportions in both octahedral and tetrahedral sites [35,36]. This different atomic arrangement in various phases of Al_2O_3 directly affects the surface properties of materials such as heat transfer rate, mean free path of phonon vibrations etc. From the literature, it is not well understood that how phases transition of Al_2O_3 NPs influence its thermo-physical properties like TC, specific heat and VS etc. So this study focuses on the preparation of most stable (α, γ) phase of Al_2O_3 and compares their influences on TC and VS as a function of NPs concentration, temperature, sonication and settling time.

Experimental

Materials and methods

Preparation and characterization of α-Al_2O_3 and γ-Al_2O_3 particles

Aluminium sulphate hydrate ($Al_2(SO_4)_3{\cdot}xH_2O$) and propanol ($CH_3CH_2CH_2OH$) were purchased from Loba Chemicals, India and used without further purification. De-ionized water was obtained using an ultra filtration system (Milli-Q, Milipore) with a measured conductivity 35 mho cm^{-1} at 25 °C. A 1 M $Al_2(SO_4)_3{\cdot}xH_2O$ solution was prepared by dissolving it in 20 ml distilled water and then added it slowly into 40 ml propanol and obtained white precipitate of $Al(OH)_3$ was dried in an oven at 70–80 °C for 6 to 8 h and calcined at 700 to 1100 °C in a muffle furnace for 3 to 4 h [37]. The formation of γ-phase takes place at 700 °C and formation of α-phase takes place at 1100 °C. Their crystal structure and grain size were investigated by powder X-ray diffraction study with PANalytical's X'Pert Pro with Cu *K*α (λ = 1.5406 Å) radiation. The size and shape analysis was done by Transmission Electron Microscope (Hitachi 7500 model) with Al_2O_3 powder samples dispersed in methanol.

Thermal conductivity, viscosity and electro-kinetic parameter analysis

Al_2O_3 particles were dispersed into DI-water with different VF in the range of 0.01% to 1%. Ultrasonication (frequency ± 42 kHz ± 6%, Bransonic-3510 E-DTH) has been carried out for several hours (2 to 10 h) to prevent particle's agglomeration. The TC of all samples was measured by using KD2 Pro Thermal Property Analyzer (Decagon Devices, Inc., USA) after calibrating with glycerin (provided by Decagon Company) and DI-water. The VS of the aqueous suspension was measured by a Brookfield Programmable Rheometer (Model: DV-III ULTRA) appropriately connected to Temperature Controlled Bath to vary the fluid temperature between 5 and 80 °C. Particle size distribution and surface charge was measured by (DLS) Dynamic Light Scattering (Brook haven 7610 instrument) and Zeta potential analyzer. Aqueous suspension of 20 ml (0.05 VF) of Al_2O_3 was sonicated for 4 to 8 h and used (1.5 ml) for DLS and zeta potential measurement.

Results and discussion

Structural and morphological characterization

The XRD pattern in Fig. 1(a) shows the strongly intense sharp peaks characteristic to rhombohedral structure of α-Al_2O_3 with R3c (1 6 7) symmetry as per JCPDS Card No: 46-1212. The sharp diffraction peaks indicative of high crystallinity of α phase whose average crystallite size was determined by the Scherrer formula [38] and found in the range of 30–40 nm for α-phase and 4–10 nm for γ-phase. No any other characteristic peaks of impurities have been observed, indicating the high purity of as-prepared sample. Fig. 1(b) shows relatively broad and weak peaks (1 1 1), (3 1 1), (4 0 0) and (4 4 0) for cubic structure of amorphous γ-Al_2O_3 phase (JCPDS Card No. 10-0425). Transformation of all pure alumina into the single α-phase was observed starting from 1050 °C, as indicated by Fig. 1a. The sharp peaks of the α-phase indicate the relatively large grain sizes and well-defined long-range order, which signifies the collapse of the porous structure characteristic of the low-temperature phases (χ, κ, γ, η, δ, θ, and ρ) and the consequent diminishment of the powder surface area [39]. The formation of γ-Al_2O_3 is due to progressive dehydration and desorption of surface hydroxyl groups that led to its amorphous nature. The peaks exhibit broad and diffuse profiles, indicating the presence of small crystalline grains and compositional fluctuations. This is consistent with the location of the Al^{3+} ions either by the tetrahedral or octahedral sites within the spinel structure [40].

The Transmission Electron Microscopy (TEM) images in Fig. 2(a and b) showed well crystalline α-Al_2O_3 NPs, which are overlapped with each other. An individual particle shows size range between 40 and 50 nm but due to overlapping of large number of particles, size increases up 0.10 μm and could not be reduced even with significant ultra-sonication. Fig. 2(c and d) displayed TEM images of amorphous γ-Al_2O_3 NPs which are also agglomerated and form large cluster. The actual size or individual size of NPs is in 30–40 nm range but it tends to agglomerate and resulted in large

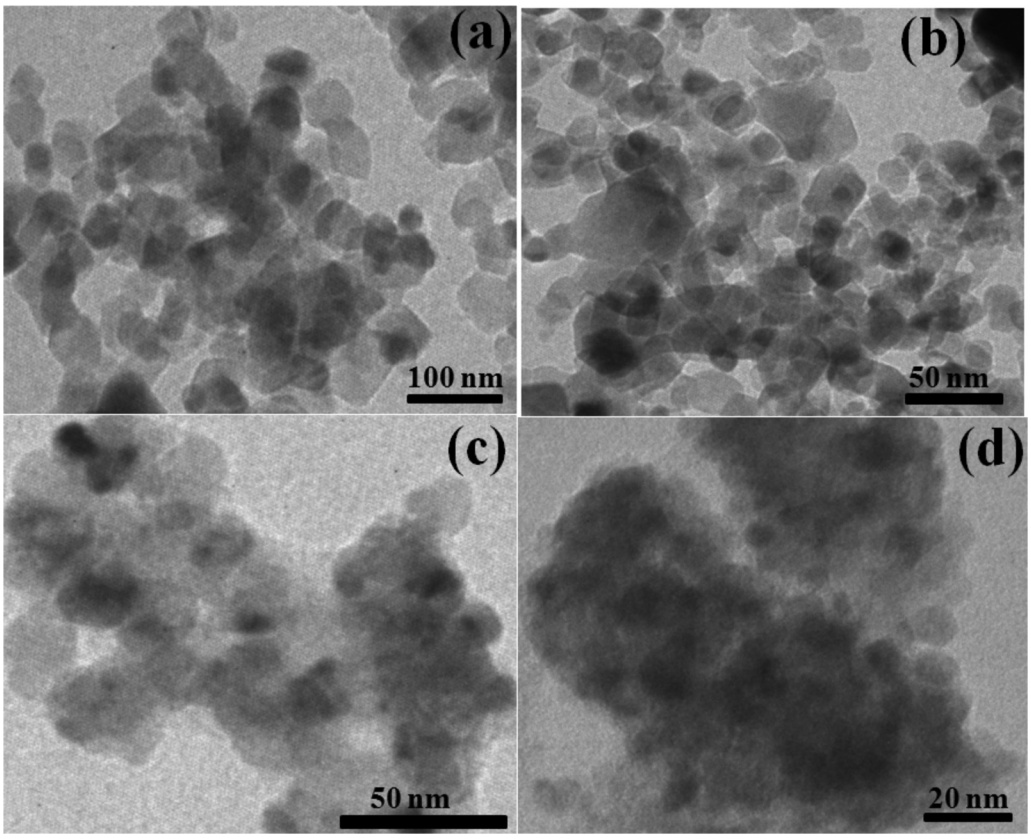


Fig. 2. TEM images of: (a and b) α-Al_2O_3 and (c and d) γ-Al_2O_3 particles.

cluster with size up to 0.15 μm. The reason for overlapping of NPs and consequence in large cluster is the more interaction between hydroxyl groups present on the surface, trapping of small particles into large cluster, high surface-to-volume ratio, high surface charge or van der Waals interaction between them [41]. The NPs size distribution in aqueous was also verified by Dynamic light scattering and zeta potential analyzer, which also shows μm size of particles.

Thermal conductivity

Fig. 3(a) shows that with an increase in sonication time, TC increases for α-Al_2O_3 and γ-Al_2O_3 suspension in comparison to DI water having TC = 0.597 W/m K at 26 °C. During 2 to 4 h sonication, the TC increases slightly, but after 6–8 h it increases sharply and decreases thereafter 10 h of sonication. The α-Al_2O_3 aqueous suspension always displayed higher TC (4–6%) than γ-Al_2O_3 aqueous suspension. It was also found that TC increases with increase in volume fraction from 0.01 to 1 vol.% for both phases, but again α-phase always shows more enhancement in TC than γ-phase as shown in Fig. 3(b). The reason for higher TC in α-Al_2O_3 aqueous suspension is its high crystallinity than amorphous γ-phase. In γ-phase some void spaces are present in its crystal structure which becomes obstacle in the path of phonon vibration and thereby decreases heat transfer rate. It is evident that there is an optimum sonication time (8 h) at which the Al_2O_3 aqueous suspension found to be more stable. This is due to supersonic waves travel longitudinally within the liquid medium and cause alternate positive and negative pressure waves in the liquid, which results in the homogeneous distribution of particles in the continuous liquid phase and Brownian motion of particles increases and helps in rapid heat transfer rate [42]. The decrease

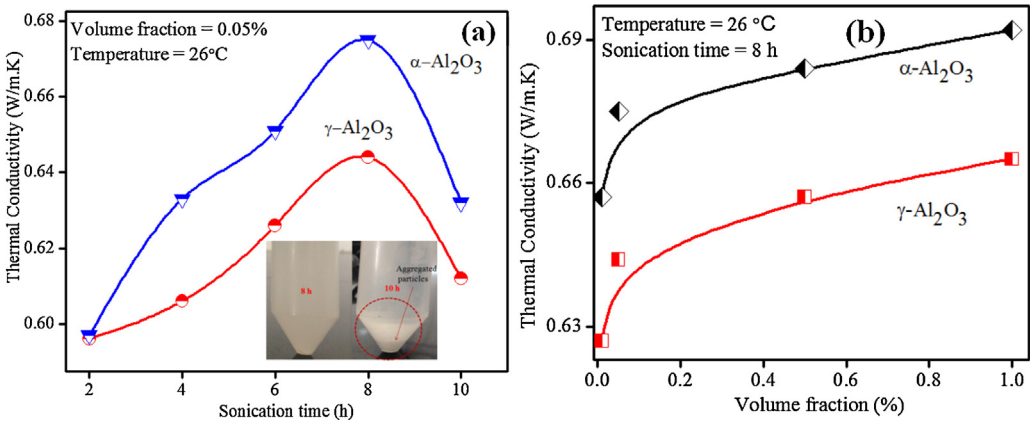


Fig. 3. Variation in thermal conductivity of Al_2O_3–DI water dispersion with (a) Sonication time and (b) Volume fraction.

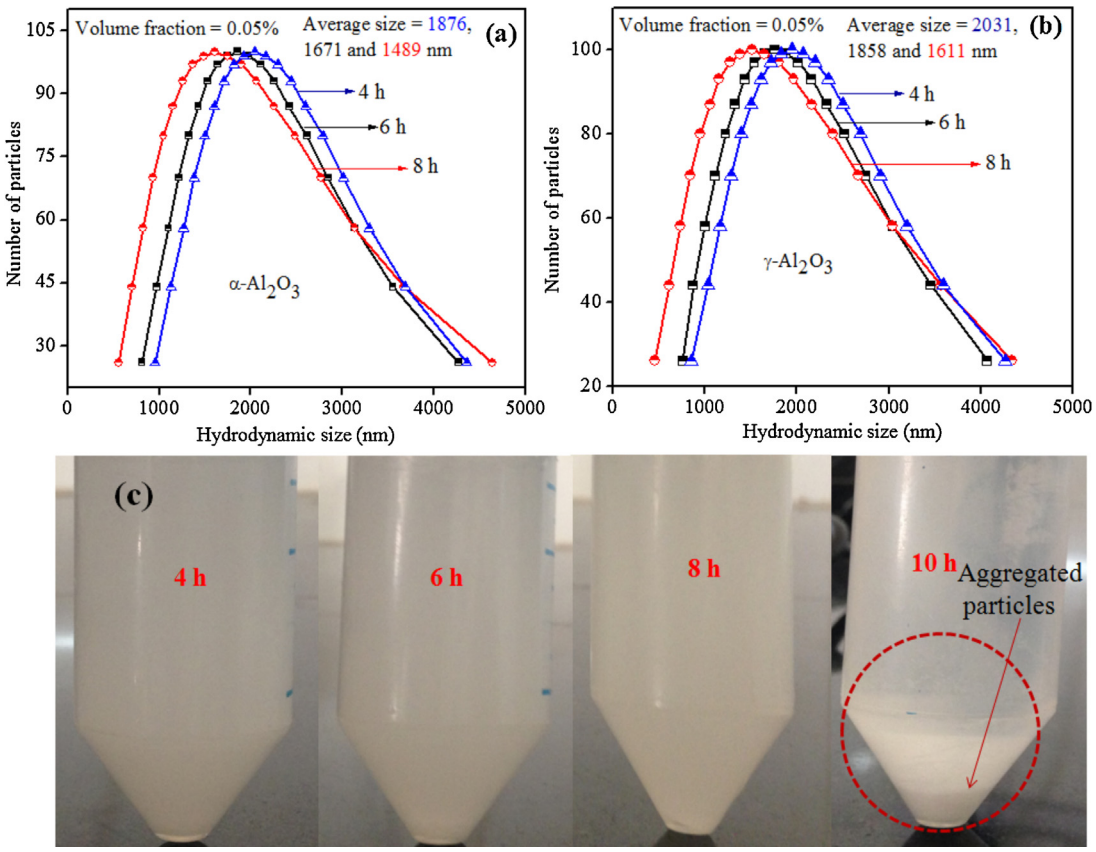


Fig. 4. Particle size distribution of (a) α-Al_2O_3, (b) γ-Al_2O_3 dispersed in de-ionized water after 4 to 8 h sonication and (c) Photographs of Al_2O_3 suspension after sonication for 4–10 h.

in stability and TC after 10 h sonication is probably due to the formation of some defects/disorder in Al_2O_3 structure as reported by some research groups [41,43,44]. The most probable explanation for increased volume fraction is that the distance between NPs decreases as the particle concentration increases and enhancement in the TC can be attributed to increased particle-to-particle interactions at higher concentrations [45], but optimization of volume fraction is also important here for commercialization of these aqueous suspension for cooling applications.

The effect of sonication time on dispersion stability was further verified by DLS particle size distribution and Zeta potential measurements. The decrease in cluster size (1.8 to 1.4 μm) with increasing the sonication time from 4 to 8 h was due to strong sonic waves which break down large cluster into smaller one as presented in Fig. 4. However, after 10 h sonication, the particle's surface charge density might be increased [41] which leads to aggregation due to higher force of attraction between them. As a result, the particles settled down as shown in Fig. 4(c), hence DLS particle size distribution do not give any conclusive result that might be beyond the upper limit of instrument's measurement (2 nm–5 μm). The improved dispersion stability was also verified by zeta potential measurement. For α-Al_2O_3 and γ-Al_2O_3 suspension (0.05%) sonicated for 4 to 8 h, the zeta potential values varies from −33 to −38 mV and −31 to −35 mV (characteristic value for stable suspension), respectively, indicating better dispersion stability than 10 h sonicated sample showing notably lower zeta potential

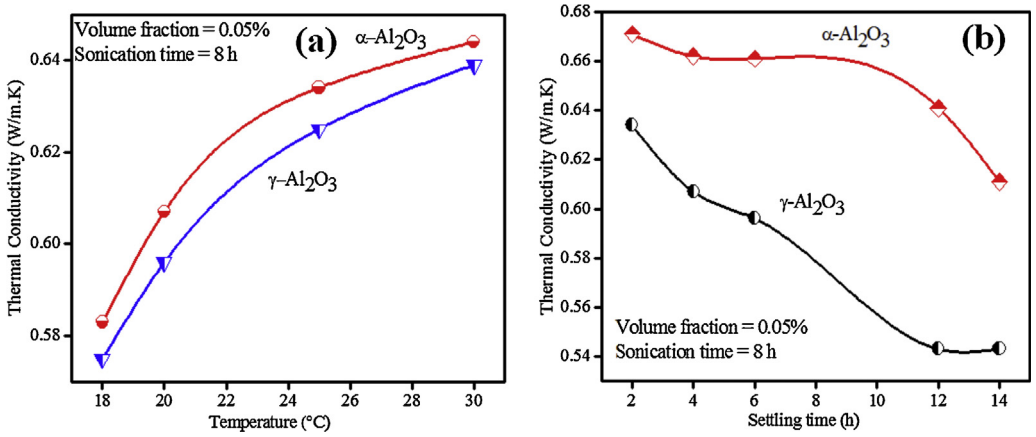


Fig. 5. Variation in thermal conductivity of Al_2O_3–DI water dispersion as a function of (a) Temperature and (b) Settling time.

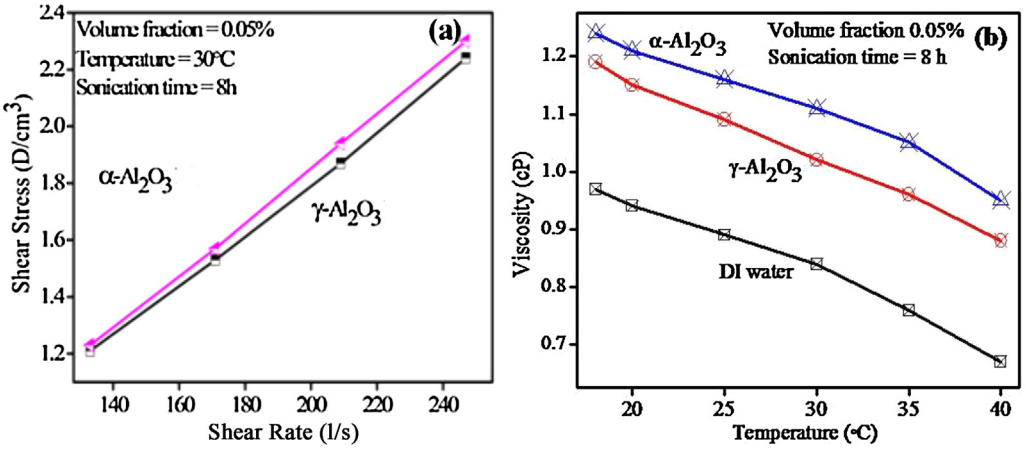


Fig. 6. (a) Shear stress as a function of shear rate and (b) Viscosity of Al_2O_3–water dispersion at different temperature.

−12 mV probably due to particles agglomeration which quickly settles down as evident in the comparative images (Fig. 4c) of Al_2O_3 suspension.

The heat-transfer rate depends on the temperature gradient and TC of the material. Change of temperature affects the Brownian motion of particles and clustering of particles, which results in dramatic changes of TC. The variation in TC for both aqueous suspensions with temperature (Fig. 5a) from 18 to 30 °C revealed that TC increases with increase in temperature, where α-Al_2O_3 showed more enhancements in TC than γ-Al_2O_3 due to more crystallinity of α-phase. The Brownian motion intensifies with an increase in temperature as per the kinetic theory of particles. With the increase in temperature, the Brownian motion and phonon vibration become stronger and this assists the rapid heat transfer rate from one particle to another.

Another aspect of aqueous suspension research that has received relatively low attention is the effect of settling time on the TC. In this paper, settling time indicates the time after the completion of sonication. Even though nano-size dispersion is expected to remain stable, this is generally not the case in reality (especially when the powders are added in solvents in a two step process). The NPs tend to agglomerate, form clusters and settle down; this reduction in the stability could decrease the TC of suspension with time [46]. Contrary to this theory, however, is another theory according to which aggregation is a more likely cause for the enhanced TC of suspension. Thus, there is no agreement between the various research groups about the relation of particle agglomeration, stability and the various thermo-physical properties of suspension. This is without doubt that the time dependent properties of suspensions are to be examined for commercial use, where long-term stability is essential factor. In the present experimental program, suspensions were prepared after 8 h of sonication and then left undisturbed for several hours. One reading was tested for TC just after the end of sonication (i.e. a freshly prepared sample). Afterwards, in every 2 h, the samples were tested for TC, i.e. the different settling timings achieved were: 0, 2, 4, 6, 12 and 14 h. It is clearly shown that during initial hours (2 to 6 h) of settling time, α-Al_2O_3 aqueous suspension showed slight decrease in TC and thereafter a sharp decrease in TC was observed (Fig. 5b) after 12 to 14 h settling, whereas γ-Al_2O_3 aqueous suspension shows sharp decrease in TC in initial hours and then become almost constant beyond 8 h settling. The reason for this is stronger attractive forces by which large size cluster formation takes place and prone to precipitating down to the bottom of the liquid. Smaller particles, in contrast, are more easily dispersed and suspended in liquid when they are affected by repulsive forces. Both van der Waals and electrostatic forces are important factors in the dispersion stability of solids in liquid.

Viscosity of Al_2O_3–DI water suspensions

Viscosity describes the internal resistance of a fluid to flow and it is an important property for all thermal applications involving fluids, the pumping power is related with the viscosity of a fluid. The VS of Al_2O_3 aqueous suspension is affected by many factors, such as volume fraction, particle size, particle aspect ratio and aggregate size [47]. In this case we address two other factors, shear-rate dependence and temperature dependence VS, which are very important for the fundamental understanding of their heat transfer applications. The shear stress increases linearly with the shear rate for low loading (0.05%) of Al_2O_3 in DI-water at 30 °C indicating the Newtonian behavior of Al_2O_3 DI-water suspension (Fig. 6a). The study of the temperature dependence of viscosity of suspension is also very important because it influences the hydrodynamics that affects the heat transfer of cooling system. Fig. 6(b) shows that viscosity reduces with an increase in temperature, indicates that with an increase in temperature the intra and intermolecular interactions between the molecules become weak and increase the surface tension which automatically becomes less viscous. The correlation between TC, VS and temperature is shown in Fig. 7 and it was found that with increase in temperature TC increases but VS decreases, so they are inversely

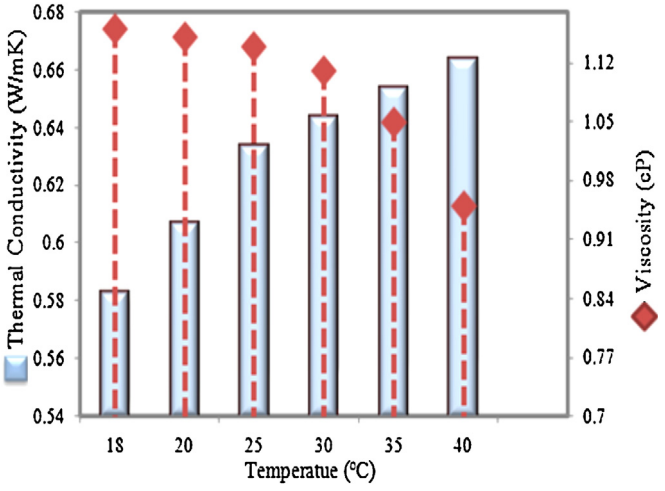


Fig. 7. Comparative graph of thermal conductivity and viscosity of α-Al_2O_3 aqueous suspension as a function of temperature.

proportional to each other. As the temperature increases, the time of interaction between neighbouring molecules of a liquid decreases because of the increased velocities of individual molecules and increases the average kinetic energy of the molecules in a liquid. The greater average kinetic energy of the molecules more easily overcomes the attractive forces that tend to hold the molecules together.

Conclusions

In summary, an experimental investigation of the TC and VS for α and γ-Al_2O_3 aqueous suspension was studied. It was found that α-phase is best for thermal conduction as compare to γ-phase due to its effective crystalline nature. The α-Al_2O_3 dispersion is more stable than γ-Al_2O_3 dispersion for longer time which directly affects the TC of suspension. The TC remarkably increases with temperature, concentration, sonication time, crystallinity and stability of NPs in aqueous suspension. The correlation between TC and VS as a function of temperature is inversely proportional to each other. From viscosity data it was found that loading of lower particle concentration, the Al_2O_3 aqueous suspension showed Newtonian behavior for both phases. So it is concluded here that optimization of experimental conditions for higher TC and long term stability of NPs in conventional heat transfer fluids is very importance to their commercial availability in cooling applications.

Acknowledgements

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